

Commit 1b17f14

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ActuallyNull committed last month

model v1.6, note: forgot to commit v1.5; (v1.5, modularised v1.4 architecture, implemented validation sets to model.fit) v1.6: implemented resnet arch, implemented sparse categorical focal loss; so far best model yet

main

1 parent 7a0765b commit 1b17f14



Filter files...



model.py

1 file changed +31 -17 lines changed

Search within code



model.py+31 -17

@@ -8,6 +8,7 @@

88from sklearn.preprocessing import label_binarize

99from sklearn.utils.class_weight import compute_class_weight

1010from keras import regularizers

11+ from focal_loss import SparseCategoricalFocalLoss

12import numpy as np

1314X_train, y_train, X_val, y_val, X_test, y_test, le = create_train_val_test_splits()

@@ -20,33 +21,45 @@

2021ReduceLROnPlateau(monitor='val_loss', factor=0.2, patience=5, min_lr=1e-6, verbose=1)

2122]

2223

- def ecg_cnn(input_length, num_classes=3): # 0: afib, 1: normal, 2: other arrhythmia

- inputs = layers.Input(shape=(input_length, 1))

-

- x = layers.Conv1D(32, kernel_size=15, padding='same')(inputs)

+ def conv_block(x, filters, kernel_size, pool_size=2, dropout=0.2):

+ x = layers.Conv1D(filters, kernel_size, padding='same')(x)

x = layers.BatchNormalization()(x)

x = layers.ReLU()(x)

- x = layers.MaxPooling1D(pool_size=2)(x)

- x = layers.Dropout(0.2)(x)

+ x = layers.MaxPooling1D(pool_size=pool_size)(x)

+ x = layers.Dropout(dropout)(x)

+ return x

3131

- x = layers.Conv1D(64, kernel_size=11, padding='same')(x)

+ def residual_block(x, filters, kernel_size=15, strides=1):

+ x_skip = x

+ x = layers.Conv1D(filters, kernel_size, strides=strides, padding='same', kernel_regularizer=regularizers.l2(1e-4))(x)

x = layers.BatchNormalization()(x)

x = layers.ReLU()(x)

- x = layers.MaxPooling1D(pool_size=2)(x)

- x = layers.Dropout(0.2)(x)

-

- x = layers.Conv1D(128, kernel_size=7, padding='same')(x)

+ x = layers.Conv1D(filters, kernel_size, strides=1, padding='same', kernel_regularizer=regularizers.l2(1e-4))(x)

3938x = layers.BatchNormalization()(x)

39+ if strides > 1:

40+ x_skip = layers.Conv1D(filters, 1, strides=strides, padding='same')(x_skip)

41+ x_skip = layers.BatchNormalization()(x_skip)

42+ x = layers.Add()([x_skip, x])

4043x = layers.ReLU()(x)

41- x = layers.MaxPooling1D(pool_size=2)(x)

42- x = layers.Dropout(0.2)(x)

44+ return x

4345

- x = layers.Conv1D(256, kernel_size=5, padding='same')(x)

+ def ecg_cnn(input_length, num_classes=3): # 0: afib, 1: normal, 2: other arrhythmia

+ inputs = layers.Input(shape=(input_length, 1))

+

+ x = layers.Conv1D(64, 15, strides=2, padding='same')(inputs)

x = layers.BatchNormalization()(x)

x = layers.ReLU()(x)

+ x = layers.MaxPooling1D(pool_size=3, strides=2, padding='same')(x)

+

+ x = residual_block(x, 64)

+ x = residual_block(x, 64)

+ x = residual_block(x, 128, strides=2)

+ x = residual_block(x, 128)

+ x = residual_block(x, 256, strides=2)

+ x = residual_block(x, 256)

+

x = layers.GlobalAveragePooling1D()(x)

x = layers.Dropout(0.2)(x)

-

x = layers.Dense(64, activation='relu', kernel_regularizer=regularizers.l2(0.001))(x)

x = layers.Dropout(0.5)(x)

outputs = layers.Dense(num_classes, activation='softmax')(x)

@@ -57,15 +70,16 @@ def ecg_cnn(input_length, num_classes=3): # 0: afib, 1: normal, 2: other arrhyth

5770model = ecg_cnn(input_length=3000) # 300 hz for 10 sec = 3000 samples

5871

5972model.compile(optimizer='adam',

- loss='sparse_categorical_crossentropy',

+ loss=SparseCategoricalFocalLoss(gamma=2),

metrics=['accuracy'])

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6376model.fit(X_train,

6477y_train,

6578epochs=100,

6679batch_size=64,

6780callbacks=callback,

- class_weight=class_weights

+ class_weight=class_weights,

+ validation_data=(X_val, y_val)

6983)

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7185test_loss, test_acc = model.evaluate(X_test, y_test)

Comments0



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